Liyan Sui

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China

University of Electronic Science and Technology of China
Supervisor: Prof. Yan Zhang (IEEE Fellow)

SUMMARY

- I am a passionate researcher focusing on exploring deep learning algorithms to solve optimization problems in edge computing networks. My work aims to apply artificial intelligence techniques to support efficient and trustworthy resource allocation in dynamic network environments.
- My research interests include edge intelligence, mobile edge computing, computing power networks, resource allocation, and privacy protection in dynamic network environments.

EDUCATION

- **Doctor of Philosophy in Information and Communication Engineering** **(Sep. 2023 - Present)**
  University of Electronic Science and Technology of China
- **Visiting Student in Electrical and Electronic Engineering** **(Nov. 2025 - Nov. 2026)**
  Nanyang Technological University Host Supervisor: Prof. Chau Yuen (IEEE Fellow)
- **Master in Cyberspace Security** **(Sep. 2020 - Jun. 2023)**
  Heilongjiang University
- **Bachelor in Information Security** **(Sep. 2015 - Jun. 2019)**
  Qingdao University

PUBLICATIONS

Journal Papers

- **L. Sui**, X. Huang, K. Zhang, Y. Zhang, F. Wu, X. Guan, S. Xu and Y. Zhang "Energy-Privacy Tradeoff for Task Matching in Edge Computing Power Networks", *IEEE Transactions on Cloud Computing (TCC)*, vol. 13, no. 4, pp. 1315-1326, Oct.-Dec. 2025. **(JCR Q1, IF: 5.3)**
- **L. Sui**, K. Zhang, F. Wu and X. Huang, "Large Models for Resource Allocation in Edge Computing Power Networks,", *IEEE Network*, vol. 39, no. 4, pp. 82-89, July 2025. **(JCR Q1, IF: 8.8)**
- **L. Sui**, X. Guan, C. Cui, H. Jiang, H. Pan and T. Ohtsuki,, "Graph Learning Empowered Situation Awareness in Internet of Energy With Graph Digital Twin", *IEEE Transactions on Industrial Informatics (TII)*, vol. 19, no. 5, pp. 7268-7277, May 2023. **(JCR Q1, IF: 9.8)**

Conference Papers

- **L. Sui**, X. Huang, and Y. Zhang, "Task Rescheduling for Service Continuity in Vehicular Computing Power Networks," *IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)* accepted.

ACADEMIC EXPERIENCE

- **Jan.2025 - Present:** Participated in the research project of the Key Laboratory for Computing Power Networks and Information Security, Ministry of Education of China: "Secure and Trustworthy Offloading in Vehicular Computing Power Networks". Focused on protecting the computational power information of edge devices while enabling energy-efficient task offloading. To address the challenges of security and efficiency in dynamic environments, a combined approach integrating graph neural networks and reinforcement learning was employed to support trustworthy and adaptive decision-making.
- **Sep. 2022 - Jun. 2023:** Participated in the research project with State Grid Corporation of China: "Market-Based Strategies for Enhancing Renewable Energy Integration under the Dual-Carbon Goals.
- **Jul. 2022 - May 2023** Participated in the research project with State Grid Corporation of China: "Low-Carbon Dispatching Strategies for New-Type Power Systems under the Dual-Carbon Goals.

WORK IN PROCESS

- **L. Sui**, X. Huang, Y. Zhang and Y. Zhang, "Gated Federated Learning for Mobility-Aware Task Matching in Vehicular Computing Power Networks," *IEEE Transactions on Intelligent Transportation Systems (TITS)* under review.
- **L. Sui**, X. Huang, J. Wu, BH. Snoog and Y. Zhang, "Generative AI Approach for Task Rescheduling in Fault-Tolerant Vehicular Computing Power Networks", *IEEE Transactions on Mobile Computing (TMC)* under review.